Lecture 1

Lecture 1 reviews pure Einstein gravity in $2+1$ dimensions, first with $\Lambda = 0$ — conical defects, spinning particles, and the upper bound on the energy of an asymptotically flat spacetime — and then with $\Lambda < 0$, where we set up AdS_3, its isometry group, and the construction of new solutions by discrete identifications. The BTZ black hole is built explicitly as a quotient, and we explain why the same trick produces no black hole at $\Lambda = 0$ but does produce higher-genus black holes in AdS_4.

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Conventions

Signature $(-,+,+)$; $\mu,\nu = 0,1,2$; $R^{\rho}{}_{\sigma\mu\nu} = \partial_\mu \Gamma^{\rho}_{\nu\sigma} - \ldots$, $R_{\mu\nu} = R^{\rho}{}_{\mu\rho\nu}$, so that the round sphere has $R > 0$. Units $c = \hbar = 1$. The action is

$$I = \frac{1}{16\pi G}\int \mathrm{d}^3x\,\sqrt{-g}\,\big(R - 2\Lambda\big) + (\text{boundary terms}), \qquad \Lambda = -\frac{1}{\ell^2}\ \ (\Lambda < 0 \text{ case}). \tag{1}$$

The dimension of G follows from the action: $[\mathrm{d}^3x] = L^3$ and $[R] = L^{-2}$, so $\frac{1}{G}\int \mathrm{d}^3x\sqrt{-g}\,R$ is dimensionless only if

$$[G] = \text{length} = \frac{1}{\text{mass}} \qquad (\hbar = c = 1), \tag{2}$$

so that GM is *dimensionless*. Contrast four dimensions, where $[G_4] = \text{length/mass}$ and $G_4 M$ is the Schwarzschild radius: in $3d$ no length can be built from G and M alone, and masses are naturally measured in units of $1/G$. Two consistency checks on (2): the central charge $c = 3\ell/2G$ of §3 (lecture 2) and the Chern–Simons level $k = \ell/4G$ of §4 (lecture 2) are pure numbers, which forces $[G] = [\ell]$. Also dimensionless are GJ/ℓ, $M\ell$, $\mathcal{L}$, and the entropy $S = \pi r_+/2G$. The Planck length appears only through $\ell_{\mathrm{P}} = \hbar G$.

Some references instead set $8G = 1$ and treat M as dimensionless; then G is a pure number but c and k carry dimensions. We do not use that convention.

Boundary lightcone coordinates are

$$x^{\pm} = \frac{t}{\ell} \pm \varphi, \qquad \varphi \simeq \varphi + 2\pi, \qquad \partial_{\pm} = \tfrac{1}{2}\Big(\ell\,\partial_t \pm \partial_\varphi\Big). \tag{3}$$

1 Gravity in three dimensions with $\Lambda = 0$

1.1 Kinematics: why $3d$ gravity has no local degrees of freedom

The starting point — and the reason the whole subject exists — is a statement of linear algebra. In D dimensions the Riemann tensor has $\frac{1}{12}D^2(D^2-1)$ independent components while the Ricci tensor has $\frac{1}{2}D(D+1)$. For $D=3$ both equal 6: the Weyl tensor vanishes identically and the Riemann tensor is determined algebraically by the Ricci tensor,

$$R_{\mu\nu\rho\sigma} = g_{\mu\rho}R_{\nu\sigma} + g_{\nu\sigma}R_{\mu\rho} - g_{\nu\rho}R_{\mu\sigma} - g_{\mu\sigma}R_{\nu\rho} - \frac{R}{2}\big(g_{\mu\rho}g_{\nu\sigma} - g_{\mu\sigma}g_{\nu\rho}\big). \tag{4}$$

Consequently the vacuum equations $G_{\mu\nu} + \Lambda g_{\mu\nu} = 0$, i.e. $R_{\mu\nu} = 2\Lambda g_{\mu\nu}$ and $R = 6\Lambda$, force

$$R_{\mu\nu\rho\sigma} = \Lambda\big(g_{\mu\rho}g_{\nu\sigma} - g_{\mu\sigma}g_{\nu\rho}\big)\,: \tag{5}$$

every solution of the vacuum equations is a space of constant curvature, hence locally isometric to Minkowski ($\Lambda = 0$), de Sitter ($\Lambda > 0$) or anti-de Sitter ($\Lambda < 0$) space. There is no gravitational radiation, and the graviton count $\frac{1}{2}D(D-3) = 0$ confirms the absence of local degrees of freedom. This is the content of the classic analysis of Deser, Jackiw and 't Hooft [7] and Deser–Jackiw [8].

The physics is therefore entirely *global*: all the information in a solution lies in the topology of the spacetime, in the holonomies of the flat (or constant-curvature) connection around non-contractible cycles, and in the boundary conditions. A modern phrasing: a solution is a (G, X)-structure, with X the model space and $G = \mathrm{ISO}(2,1)$, $\mathrm{SO}(3,1)$ or $\mathrm{SO}(2,2)$ its isometry group, and the moduli space of solutions on a spatial surface Σ is

$$\mathcal{M} \;\subseteq\; \mathrm{Hom}\big(\pi_1(\Sigma), G\big)/G, \tag{6}$$

a finite-dimensional space [15, 18].

Exercise 1. Verify (4) by counting components and matching the symmetries, then contract it to recover $R_{\mu\nu}$ and R. Deduce (5) and conclude that the Weyl tensor vanishes identically for $D = 3$.

1.2 Point sources: conical defects

Since the vacuum is locally trivial, matter can only act globally. Take a static point particle of mass m sitting at the origin, $T^{00} = m\,\delta^{(2)}(\vec{x})$, all other components zero. Solving the linearised equations and then noting that the result is exact because the exterior is flat, one finds the celebrated conical geometry [6, 7]

$$\mathrm{d}s^2 = -\mathrm{d}t^2 + \mathrm{d}r^2 + (1-4Gm)^2 r^2\,\mathrm{d}\varphi^2, \qquad \varphi \simeq \varphi + 2\pi. \tag{7}$$

Rescaling $\tilde{\varphi} = (1-4Gm)\varphi$ makes the metric manifestly Minkowskian, but with $\tilde{\varphi}$ identified with period

$$2\pi(1-4Gm) = 2\pi - \delta, \qquad \boxed{\delta = 8\pi G m}. \tag{8}$$

The spacetime is a cone: flat everywhere except at the tip, where the curvature is a delta function of total strength δ, the *deficit angle*. Equivalently: the holonomy of the flat connection around the particle is a rotation by δ, an element of the Lorentz subgroup of $\mathrm{ISO}(2,1)$. A test particle passing the source is not attracted, but two geodesics on opposite sides are relatively deflected by δ — a purely topological "gravitational lens".

Two immediate consequences:

- Newtonian gravity has no $3d$ analogue: the "force" vanishes, but the global structure is deformed.
- Since δ is an angle, m is bounded (below, and — as we will see — above).

1.3 Spinning particles and closed timelike curves

Adding spin J (i.e. $T^{0i} = -\frac{1}{2}\varepsilon^{ij}\partial_j\big(J\delta^{(2)}(\vec{x})\big)$) produces [7]

$$\mathrm{d}s^2 = -\big(\mathrm{d}t + 4GJ\,\mathrm{d}\varphi\big)^2 + \mathrm{d}r^2 + (1-4Gm)^2 r^2\,\mathrm{d}\varphi^2. \tag{9}$$

The geometry is still flat; the holonomy is now a rotation *combined with a timelike translation* — a "helical" structure in which going once around the particle shifts t by $-8\pi GJ$. The curve $r = \text{const}$, $t = \text{const}$ has $\|\partial_\varphi\|^2 = -16G^2J^2 + (1-4Gm)^2r^2$, which is negative for

$$r < r_{\mathrm{CTC}} \equiv \frac{4G|J|}{|1-4Gm|}. \tag{10}$$

Inside r_{CTC} the orbits of ∂_φ are closed timelike curves. This is our first encounter with a phenomenon that will recur for the rotating BTZ black hole: in $3d$ the pathology of a "singularity" is typically a breakdown of *causal* structure, not of curvature.

Exercise 2. Compute the holonomy of (9) around the source explicitly as an element of $\mathrm{ISO}(2,1)$, and identify the rotation angle and the translation vector. Show that for $J \neq 0$ the holonomy has no fixed point in Minkowski space — the "particle" is a *screw motion*, not a rotation.

1.4 Multi-particle solutions, the total deficit, and the upper bound on the energy

Consider N static particles. Because the geometry is flat outside the sources, the whole spatial slice is a flat polygon with identifications; the deficits add, and asymptotically one has a single cone with total deficit

$$\delta_{\mathrm{tot}} = 8\pi G\sum_{i=1}^{N} m_i \equiv 8\pi GM. \tag{11}$$

Two thresholds appear [7, 9]:

(i) $\delta_{\mathrm{tot}} = 2\pi$, i.e. $M = \dfrac{1}{4G}$. At this value the asymptotic cone degenerates: the circle at infinity has zero opening angle and the spatial geometry becomes a semi-infinite cylinder rather than a cone. For $\delta_{\mathrm{tot}} > 2\pi$ the space closes off at finite distance and is no longer asymptotically flat.

(ii) $\delta_{\mathrm{tot}} = 4\pi$, i.e. $M = \dfrac{1}{2G}$: a *closed* universe. This is nothing but Gauss–Bonnet, $\int_\Sigma \sqrt{h}\,{}^{(2)}R = 4\pi\chi(\Sigma)$, applied to $\Sigma = S^2$ where $\chi = 2$; the total curvature is fixed by topology and hence so is the total mass.

Thus $3d$ **asymptotically flat gravity has a maximum energy**. The statement can be sharpened well beyond the point-particle argument. Ashtekar and Varadarajan [11] performed a Hamiltonian analysis of $2+1$ gravity coupled to matter satisfying the dominant energy condition in

the asymptotically flat sector and proved that the ADM Hamiltonian is bounded *both* below and above:

$$0 \;\le\; E \;\le\; \frac{1}{4G}, \tag{12}$$

with $E = 0$ if and only if the spacetime is globally Minkowski. The upper bound is nonperturbative and has no analogue in $D \ge 4$; it comes from the fact that the total energy is a boundary term measuring an *angle*, and angles are bounded. Ashtekar and Varadarajan also point out the $4d$ corollary: a $3+1$ vacuum spacetime with a spacelike Killing vector reduces to $2+1$ gravity with matter, so (12) bounds the energy per unit length of cylindrical gravitational waves.

Where is the zero of energy? The bound (12) assumes a particular normalisation. Marolf and Patiño [12] showed that if one insists on the Einstein–Hilbert action *with* the Gibbons–Hawking term and a well-posed variational principle, $2+1$ Minkowski space is assigned $E = -1/4G$, so the physical range becomes $-1/4G \le E \le 0$. The *width* of the allowed band, $1/4G$, is convention-independent; its location is not. We will see the $\Lambda < 0$ counterpart of exactly this shift when we meet the Casimir energy $M_{\mathrm{AdS}} = -1/8G$ of global AdS_3.

Remark 1. The Deser–Jackiw–'t Hooft bound is often attributed loosely to "Teitelboim" or "Henneaux". The canonical/Hamiltonian treatment of energy, angular momentum and supercharge in $2+1$ gravity and supergravity is due to Henneaux [13]; the general Regge–Teitelboim machinery for surface charges used throughout these notes is [14]. The *sharp* statement of the upper bound in the asymptotically flat sector that we quote is [11].

Exercise 3. Two identical particles of mass m at rest. Show that the exterior geometry is a cone of deficit $16\pi G m$ and that, boosting one of them, the total deficit is $8\pi G(E_1 + E_2)$ with E_i the energies rather than the masses. Deduce that a sufficiently energetic two-particle collision reaches $\delta_{\mathrm{tot}} = 2\pi$: in $3d$, "black hole formation" at $\Lambda = 0$ is replaced by "closing up the universe". (This is the Gott-pair analysis; see [10, 7].)

2 Turning on $\Lambda < 0$: AdS_3 and its quotients

2.1 AdS_3 as a hyperboloid and as a group manifold

Anti-de Sitter space in three dimensions is the universal cover of the hyperboloid

$$-X_0^2 - X_3^2 + X_1^2 + X_2^2 = -\ell^2 \tag{13}$$

embedded in $\mathbb{R}^{2,2}$ with $\mathrm{d}s^2 = -\mathrm{d}X_0^2 - \mathrm{d}X_3^2 + \mathrm{d}X_1^2 + \mathrm{d}X_2^2$. The scale ℓ appearing here is fixed by the cosmological constant: specialising (5) to $\Lambda < 0$,

$$\boxed{\Lambda = -\frac{1}{\ell^2}} \qquad \Longrightarrow \qquad R_{\mu\nu} = 2\Lambda g_{\mu\nu} = -\frac{2}{\ell^2}\, g_{\mu\nu}, \qquad R = 6\Lambda = -\frac{6}{\ell^2}, \tag{14}$$

and hence $R_{\mu\nu\rho\sigma} = -\ell^{-2}\big(g_{\mu\rho}g_{\nu\sigma} - g_{\mu\sigma}g_{\nu\rho}\big)$. So ℓ is the radius of curvature of AdS_3, and — since G is dimensionless in the sense of §0 — it is the only length scale in the classical vacuum theory. Solving (13) with $X_0 = \sqrt{\ell^2 + r^2}\cos(t/\ell)$, $X_3 = \sqrt{\ell^2 + r^2}\sin(t/\ell)$, $X_1 = r\cos\varphi$, $X_2 = r\sin\varphi$ gives *global* coordinates

$$\mathrm{d}s^2 = -\Big(1 + \frac{r^2}{\ell^2}\Big)\mathrm{d}t^2 + \frac{\mathrm{d}r^2}{1 + r^2/\ell^2} + r^2\,\mathrm{d}\varphi^2, \tag{15}$$

Note that t/ℓ is periodic with period 2π on the hyperboloid (13) itself, which therefore contains closed timelike curves. Anti-de Sitter space proper is its **universal cover**, obtained by unwrapping that time circle to $t \in \mathbb{R}$:

$$\boxed{\widetilde{\mathrm{AdS}_3} \;\equiv\; \text{universal cover of the hyperboloid (13)}, \qquad t \in \mathbb{R}.} \tag{16}$$

Notation. We write $\widetilde{\mathrm{AdS}_3}$ whenever the distinction between the hyperboloid and its cover is being used — as in (20) below, where it matters that the quotient is taken of the simply connected space — and plain AdS_3 otherwise. Both denote the same manifold (16); the tilde is a reminder, not a different object.

The key structural fact is that (13) is the statement $\det g = 1$ for the 2×2 real matrix

$$g = \frac{1}{\ell}\begin{pmatrix} X_0 + X_1 & X_2 + X_3 \\ X_2 - X_3 & X_0 - X_1 \end{pmatrix} \in \mathrm{SL}(2,\mathbb{R}), \qquad \det g = \frac{X_0^2 - X_1^2 - X_2^2 + X_3^2}{\ell^2} = 1. \tag{17}$$

The induced metric is (up to normalisation) the Killing form, $\mathrm{d}s^2 = \frac{\ell^2}{2}\mathrm{Tr}\big[(g^{-1}\mathrm{d}g)^2\big]$, so AdS_3 *is the group manifold* $\mathrm{SL}(2,\mathbb{R})$. Isometries act by left and right multiplication,

$$g \;\longmapsto\; g_L\, g\, g_R^{-1}, \qquad (g_L, g_R) \in \mathrm{SL}(2,\mathbb{R})_L \times \mathrm{SL}(2,\mathbb{R})_R, \tag{18}$$

with (g_L, g_R) and $(-g_L, -g_R)$ acting identically. Hence

$$\boxed{\mathrm{Isom}(\mathrm{AdS}_3) \;\cong\; \mathrm{SO}(2,2) \;\cong\; \big(\mathrm{SL}(2,\mathbb{R})_L \times \mathrm{SL}(2,\mathbb{R})_R\big)/\mathbb{Z}_2} \tag{19}$$

a six-dimensional group — the maximal number for a $3d$ spacetime. The two chiral factors are the origin of everything holomorphic in $\mathrm{AdS}_3/\mathrm{CFT}_2$.

At the algebra level, $\mathfrak{so}(2,2) \cong \mathfrak{sl}(2,\mathbb{R}) \oplus \mathfrak{sl}(2,\mathbb{R})$ is generated by $J_{ab} = X_a\partial_b - X_b\partial_a$ (indices raised with the $\mathbb{R}^{2,2}$ metric). In the CFT-adapted basis the two $\mathfrak{sl}(2,\mathbb{R})$'s are $\{L_{-1}, L_0, L_{+1}\}$ and $\{\bar{L}_{-1}, \bar{L}_0, \bar{L}_{+1}\}$; global AdS_3 is the unique solution invariant under all six, and this will reappear below as the statement that the vacuum has $L_0 = \bar{L}_0 = -c/24$ in the unshifted normalisation.

Exercise 4. Show that J_{03} generates t-translations and J_{12} generates φ-rotations in (15). Write ∂_t and ∂_φ in terms of $L_0, \bar{L}_0$ and verify $\ell\,\partial_t = L_0 + \bar{L}_0$, $\partial_\varphi = L_0 - \bar{L}_0$ up to normalisation.

2.2 New solutions by identification

Because all solutions are locally AdS_3, the only way to produce genuinely new global geometries in the vacuum theory is to *quotient*:

$$\mathcal{M} = \widetilde{\mathrm{AdS}_3}/\Gamma, \qquad \text{where } \Gamma \text{ is a } \textit{discrete subgroup} \text{ of the isometry group } \mathrm{SO}(2,2), \tag{20}$$

with $\widetilde{\mathrm{AdS}_3}$ as in (16): one quotients the *simply connected* space, so that Γ is the fundamental group $\pi_1(\mathcal{M})$ of the result. The simplest and most important case is $\Gamma = \{e^{2\pi n\,\xi}\} \cong \mathbb{Z}$, generated by the exponential of a single Killing vector $\xi \in \mathfrak{so}(2,2)$.

Remark 2 (The isometry group on the cover)**.** The isometry group of $\widetilde{\mathrm{AdS}_3}$ is the corresponding cover of $\mathrm{SO}(2,2)$, not $\mathrm{SO}(2,2)$ itself: on the cover the time translation $t \to t + 2\pi\ell$ is a non-trivial isometry, whereas on the hyperboloid it is the identity. The Lie algebra is $\mathfrak{so}(2,2)$ either way, so nothing in the local analysis changes, but the *elliptic* conjugacy classes are affected — a rotation by $2\pi n$ is trivial in $\mathrm{SO}(2,2)$ but not upstairs, which is exactly what allows the multiply-wound conical geometries of §2.3 (lecture 2). We will not need this distinction again in Week 1.

This is the construction of Bañados, Henneaux, Teitelboim and Zanelli [16].
Two conditions must be imposed for the quotient to be a sensible spacetime.

(a) **Freeness / properly discontinuous action.** Γ must act without fixed points, otherwise the quotient has conical singularities (which we may nevertheless choose to allow — see the conical defects below).

(b) **Causal regularity.** If ξ is timelike somewhere, $\xi\cdot\xi<0$, then the identification $P\sim e^{2\pi\xi}P$ creates closed timelike curves there. BHTZ prescribe removing the region $\xi\cdot\xi\le 0$; the surface $\xi\cdot\xi=0$ becomes a *singularity in the causal structure* of the quotient even though the curvature is perfectly finite.

How ξ is encoded in (g_L,g_R)

Because $\mathrm{AdS}_3=\mathrm{SL}(2,\mathbb{R})$ and isometries act as $g\mapsto g_L\,g\,g_R^{-1}$, an infinitesimal isometry is a *pair* of $\mathfrak{sl}(2,\mathbb{R})$ elements. Writing $g_L=e^{s\lambda_L}$, $g_R=e^{s\lambda_R}$ and differentiating at $s=0$,

$$\boxed{\delta g=\xi(g)=\lambda_L\,g-g\,\lambda_R,\qquad (\lambda_L,\lambda_R)\in\mathfrak{sl}(2,\mathbb{R})\oplus\mathfrak{sl}(2,\mathbb{R})\cong\mathfrak{so}(2,2).}\tag{21}$$

So ξ *is* the pair (λ_L,λ_R): the "left" and "right" halves are exactly the two chiral $\mathfrak{sl}(2,\mathbb{R})$ factors, which asymptotically become the two Virasoro algebras of §3 (lecture 2). The norm of ξ is controlled by the two Casimirs,

$$\xi\cdot\xi=\tfrac{\ell^2}{2}\Big[\mathrm{Tr}\lambda_L^2+\mathrm{Tr}\lambda_R^2\Big]-\ell^2\,\mathrm{Tr}\big[g^{-1}\lambda_L\,g\,\lambda_R\big],\tag{22}$$

whose *first* (constant) piece is fixed by the conjugacy classes of $\lambda_{L,R}$ while the *second* depends on the point g — this is why $\xi\cdot\xi$ can change sign, and hence why the quotient can develop a causal boundary.

The finite identification $\Gamma=\{(g_L,g_R)^n\}$ is therefore labelled by the conjugacy class of g_L and of g_R *separately*. In $\mathrm{SL}(2,\mathbb{R})$ these are *elliptic* ($|\mathrm{Tr}|<2$, a rotation), *parabolic* ($|\mathrm{Tr}|=2$, $g\neq\pm\mathbf{1}$, a null rotation) and *hyperbolic* ($|\mathrm{Tr}|>2$, a boost). Equivalently, in the vector language $\xi=\frac{1}{2}\omega^{ab}J_{ab}$, the two invariants are $\frac{1}{2}\omega_{ab}\omega^{ab}$ and $\frac{1}{8}\varepsilon_{abcd}\omega^{ab}\omega^{cd}$. This gives the complete classification [16, 17]:

(g_L,g_R)	quotient	$\mathcal{L},\bar{\mathcal{L}}$ (see §2.2 (lecture 2))
elliptic × elliptic	conical defect / spinning particle	$\mathcal{L},\bar{\mathcal{L}}<0$
parabolic × parabolic	massless BTZ ($M=J=0$)	$\mathcal{L}=\bar{\mathcal{L}}=0$
hyperbolic × hyperbolic	non-extremal BTZ	$\mathcal{L},\bar{\mathcal{L}}>0$
hyperbolic × parabolic	extremal BTZ ($\lvert J\rvert=M\ell$)	$\mathcal{L}>0$, $\bar{\mathcal{L}}=0$

The elliptic–elliptic case with identification angle $2\pi\alpha$ yields the AdS_3 conical defect

$$\mathrm{d}s^2=-\Big(\alpha^2+\frac{r^2}{\ell^2}\Big)\mathrm{d}t^2+\frac{\mathrm{d}r^2}{\alpha^2+r^2/\ell^2}+r^2\mathrm{d}\varphi^2,\qquad \delta=2\pi(1-\alpha),\tag{23}$$

which for $\alpha=1$ is global AdS_3 and for $\alpha\to 0$ degenerates to the $M=0$ geometry. As we will see in §2.3 (lecture 2), (23) is literally the BTZ metric analytically continued to imaginary horizon radius, $r_+=i\alpha\ell$.

Exercise 5. Show that (23) has $M=-\alpha^2/8G$, $J=0$ in the conventions of §1 (lecture 2), and that $\alpha=1$ reproduces (15). Which range of α corresponds to a deficit, and which to an excess?

2.3 Explicit construction of the BTZ black hole as a quotient

Let us carry out the hyperbolic–hyperbolic case in full; this is the BTZ black hole, and doing it explicitly makes every later statement about horizons and CTCs transparent.

Step 1: adapted coordinates on AdS_3. Split the four embedding coordinates into the two "timelike–spacelike" pairs (X_0, X_1) and (X_3, X_2) and use boost-adapted coordinates:

$$\boxed{\begin{aligned} X_0 &= \ell \cosh\mu \, \cosh\alpha, & X_1 &= \ell \cosh\mu \, \sinh\alpha, \\ X_2 &= \ell \sinh\mu \, \cosh\beta, & X_3 &= \ell \sinh\mu \, \sinh\beta. \end{aligned}} \tag{24}$$

The constraint (13) is satisfied identically: $-X_0^2 - X_3^2 + X_1^2 + X_2^2 = -\ell^2\cosh^2\mu + \ell^2\sinh^2\mu = -\ell^2$. [checked] The induced metric is

$$\mathrm{d}s^2 = \ell^2 \mathrm{d}\mu^2 + \ell^2 \cosh^2\mu \, \mathrm{d}\alpha^2 - \ell^2 \sinh^2\mu \, \mathrm{d}\beta^2. \tag{25}$$

Note ∂_α is a boost in the (X_0, X_1) plane and ∂_β a boost in the (X_3, X_2) plane; both are *hyperbolic*.

Step 2: choose the identification vector. Take

$$\xi = \frac{r_+}{\ell}\,\partial_\alpha + \frac{r_-}{\ell}\,\partial_\beta = \frac{r_+}{\ell}\big(X_0\partial_1 + X_1\partial_0\big) + \frac{r_-}{\ell}\big(X_2\partial_3 + X_3\partial_2\big), \tag{26}$$

a linear combination of two commuting boosts, with $r_\pm$ two parameters. Its norm follows from (25):

$$\xi\cdot\xi = r_+^2\cosh^2\mu - r_-^2\sinh^2\mu = r_+^2 + (r_+^2 - r_-^2)\sinh^2\mu \;\equiv\; r^2, \tag{27}$$

which *defines* the radial coordinate r. Because $\xi\cdot\xi$ is a scalar, this is a geometric definition.

Step 3: adapted spacetime coordinates. Introduce (t, φ) by

$$\alpha = \frac{1}{\ell}\Big(r_+\varphi - \frac{r_-}{\ell}t\Big), \qquad \beta = \frac{1}{\ell}\Big(r_-\varphi - \frac{r_+}{\ell}t\Big), \tag{28}$$

chosen precisely so that $\xi = \partial_\varphi$. Substituting (28) and (27) into (25) gives [checked symbolically]

$$g_{\varphi\varphi} = r^2, \qquad g_{t\varphi} = -\frac{r_+ r_-}{\ell}, \qquad g_{tt} = \frac{r_+^2 + r_-^2 - r^2}{\ell^2}, \qquad g_{rr} = \frac{1}{f(r)}, \tag{29}$$

i.e. *exactly* the BTZ metric (4)–(5) of lecture 2.

Step 4: the identification. Declaring $\varphi \simeq \varphi + 2\pi$ is the quotient by $\Gamma = \{e^{2\pi n\xi}\}$. In group language: From (17) and (24),

$$g = \begin{pmatrix} \cosh\mu\, e^{\alpha} & \sinh\mu\, e^{\beta} \\ \sinh\mu\, e^{-\beta} & \cosh\mu\, e^{-\alpha} \end{pmatrix} = e^{\frac{\alpha+\beta}{2}\sigma_3}\, e^{\mu\sigma_1}\, e^{\frac{\alpha-\beta}{2}\sigma_3}, \qquad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},\ \sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \tag{30}$$

[checked] and, using (28),

$$\frac{\alpha+\beta}{2} = -\frac{r_+ + r_-}{2\ell}\,x^-, \qquad \frac{\alpha-\beta}{2} = +\frac{r_+ - r_-}{2\ell}\,x^+. \tag{31}$$

Under $\varphi \to \varphi + 2\pi$ we have $x^+ \to x^+ + 2\pi$, $x^- \to x^- - 2\pi$, so $g \mapsto g_L\, g\, g_R^{-1}$ with

$$\boxed{g_L = \exp\Big[\frac{\pi(r_+ + r_-)}{\ell}\,\sigma_3\Big] = e^{2\pi\sqrt{\mathcal{L}}\,\sigma_3}, \qquad g_R = \exp\Big[-\frac{\pi(r_+ - r_-)}{\ell}\,\sigma_3\Big] = e^{-2\pi\sqrt{\mathcal{L}}\,\sigma_3},} \tag{32}$$

where $\sqrt{\mathcal{L}} = (r_+ + r_-)/2\ell$ and $\sqrt{\bar{\mathcal{L}}} = (r_+ - r_-)/2\ell$ anticipate §2.2 (lecture 2). Both g_L and g_R have $|\mathrm{Tr}| = 2\cosh(\dots) > 2$: **hyperbolic** × **hyperbolic**, as promised by the table. Equation (32) also makes the whole classification manifest: $\mathcal{L} < 0$ makes $\sqrt{\mathcal{L}}$ imaginary and g_L *elliptic* (conical defects), while $\mathcal{L} = 0$ degenerates g_L to a *parabolic* element (massless/extremal cases).

Step 5: where the horizon comes from. By (27), $\xi\cdot\xi = r^2 \to \infty$ at the conformal boundary and $\xi\cdot\xi \to 0$ at $r = 0$; continuing past $r^2 = 0$ makes the *closed* orbits of ξ timelike. Excising $r^2 < 0$ leaves a region whose causal past, as seen from the timelike conformal boundary, is bounded by the null surface $r = r_+$. That null surface is the event horizon.

2.4 Why this fails at $\Lambda = 0$, and what happens in $4d$

It is worth asking why the same trick does not manufacture a black hole in flat $3d$ gravity. The question deserves a concrete answer rather than a slogan, because a naive causal-structure argument gives the wrong one: asymptotically flat spacetimes in $3+1$ dimensions have perfectly good black holes, so "$\mathcal{I}$ is null" cannot by itself be the obstruction.

Take the limit and look at the metric. Send $\ell \to \infty$ in the BTZ family (4) (lecture 2) at fixed M, J:

$$f_{\text{flat}}(r) = -8GM + \frac{16G^2J^2}{r^2}, \qquad N^\varphi = -\frac{4GJ}{r^2}. \tag{33}$$

[checked symbolically: the resulting metric has $R_{\mu\nu} = 0$ identically, so it is a genuine solution of the $\Lambda = 0$ theory — it is the $\ell \to \infty$ limit of the quotient construction, known as a *flat space cosmology* [4, 5].]

Everything now turns on the *sign of* f *at large* r, because in the corotating frame $f = -g_{tt}$ and $f = 1/g_{rr}$:

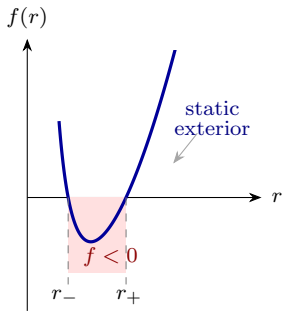
$$f > 0 \iff \begin{array}{l}\partial_t \text{ timelike, } \partial_r \text{ spacelike:}\\ \text{a } \textit{static} \text{ region}\end{array} \qquad\qquad f < 0 \iff \begin{array}{l}r \text{ is the } \textit{time} \text{ coordinate:}\\ \text{an expanding/contracting region}\end{array} \tag{34}$$

Compare the two cases (Fig. 1):

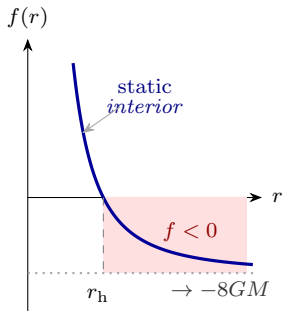
- $\Lambda < 0$**:** $f \to r^2/\ell^2 \to +\infty$. The cosmological constant *guarantees* a static region at large r, whatever M and J are. For $M > 0$ the function dips below zero in the middle and comes back up, so $\{f < 0\}$ is a finite annulus with two roots $r_\pm$. Its outer edge r_+ separates a bounded interior from an unbounded static exterior: that is what makes it an *event* horizon.
- $\Lambda = 0$**:** $f \to -8GM$. For $M > 0$ this is *negative*. There is no static region at large r at all — r is a time coordinate out there and the geometry expands. The single root, at

$$r_{\text{h}}^2 = \frac{2GJ^2}{M} \qquad \text{[checked]}, \tag{35}$$

bounds a static region *inside* it. The causal structure is therefore *inverted* relative to BTZ: r_{h} is the cosmological horizon of a static patch, not the event horizon of a black hole. Nothing is hidden; rather, everything is eventually swept up in the expansion.



(a) $\Lambda<0$, $M>0$: $f\to+\infty$
$\{f<0\}$ bounded $\Rightarrow$ *event* horizon at r_+



(b) $\Lambda=0$, $M>0$: $f\to-8GM<0$
$\{f<0\}$ unbounded $\Rightarrow$ *cosmological* horizon

Figure 1: The metric function $f(r)$ for the rotating solution, with $\Lambda<0$ (left) and $\Lambda=0$ (right); shading marks $f<0$, where r is a timelike coordinate. With $\Lambda<0$ the r^2/ℓ^2 term forces $f\to+\infty$, so the non-static region is a bounded annulus $r_-<r<r_+$ with an unbounded static exterior beyond it — an event horizon. With $\Lambda=0$, $f\to-8GM<0$ for $M>0$: the non-static region is unbounded, there is no asymptotic static observer, and the single root $r_{\rm h}$ is instead the cosmological horizon of the static region *inside* it. Curves plotted with $\ell=1$, $8GM=2$, $16G^2J^2=\frac{1}{2}$ (left) and $8GM=1$, $16G^2J^2=1$ (right).

So the r^2/ℓ^2 term is not a detail. It is the sole reason the set $\{f<0\}$ can be *bounded*, and a bounded trapped region with an unbounded static exterior is exactly what an event horizon requires.

This is the flat-space energy bound in disguise. The restriction is not new information: it is §1.4 rewritten. Match the flat conical defect (7) to the ADM form (4) (lecture 2). Setting $R=(1-4Gm)r$ turns (7) into $-\mathrm{d}t^2+\mathrm{d}R^2/(1-4Gm)^2+R^2\mathrm{d}\varphi^2$, so (after the harmless rescaling $t\to(1-4Gm)t$) it has $f=(1-4Gm)^2$, and comparing with (33) at $J=0$,

$$\boxed{8GM=-(1-4Gm)^2}\qquad\text{[checked]}\tag{36}$$

Reading off the two endpoints of the Deser–Jackiw–'t Hooft / Ashtekar–Varadarajan range $0\le m\le 1/4G$:

flat-space configuration	deficit $\delta=8\pi Gm$	$8GM$
Minkowski$_3$, $m=0$	0	-1
generic conical defect	$0<\delta<2\pi$	$-1<8GM<0$
maximal defect, $m=1/4G$	2π	0

So the *entire* physically allowed range of the $\Lambda=0$ theory is $-1\le 8GM\le 0$ — precisely the conical-defect segment of Fig. 3 (lecture 2), with Minkowski$_3$ sitting at the same point $8GM=-1$ that global AdS$_3$ occupies. The black-hole branch $8GM>0$ lies *beyond* the flat-space bound: at $\Lambda=0$ those parameters do not describe a black hole but an over-deficit configuration, which is the cosmology of Fig. 1(b). Switching on $\Lambda<0$ re-opens an asymptotic static region and turns that same branch into genuine black holes.

The construction *does* generalise upward in dimension. Since the only ingredients are (i) a maximally symmetric spacetime and (ii) a discrete subgroup of its isometry group acting properly discontinuously with a suitable causal structure, one can repeat the BTZ construction in AdS_4 and higher. This was done by the Stockholm group — Åminneborg, Bengtsson, Holst and Peldán, *Making anti-de Sitter black holes* [1] — and the outcome is genuinely new: the quotients are locally AdS_4, have event horizons, and the horizons are *tori or higher-genus Riemann surfaces* rather than spheres. They may have one or two asymptotic regions. Such topologies are forbidden for stationary black holes in $3+1$ asymptotically *flat* vacuum gravity by Hawking's horizon topology theorem — the theorem's proof uses asymptotic flatness, and it is exactly that hypothesis which the AdS boundary conditions relax. Follow-ups: Holst–Peldán [2] on the causal structure of these spacetimes, and Bañados [3] on constant-curvature black holes in general dimension. This will be a natural place to return to when we discuss higher-genus quotients and Euclidean wormholes.

Exercise 6. Verify (22) by writing $\xi\cdot\xi = \frac{\ell^2}{2}\mathrm{Tr}\big[(g^{-1}\lambda_L g - \lambda_R)^2\big]$ using $\mathrm{d}s^2 = \frac{\ell^2}{2}\mathrm{Tr}\big[(g^{-1}\mathrm{d}g)^2\big]$. Then evaluate it for the BTZ generator $\lambda_L = \sqrt{\mathcal{L}}\,\sigma_3$, $\lambda_R = -\sqrt{\mathcal{L}}\,\sigma_3$ on (30) and recover $\xi\cdot\xi = r^2$.

Exercise 7. Repeat Steps 1–4 for the elliptic–elliptic class, i.e. replace the two boosts in (26) by two rotations, and derive (23) directly. Where does the fixed point of Γ — the tip of the cone — sit in embedding coordinates?

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